

Hybrid LSTM-XGBoost Model with Optimised Transfer Learning for Residential Load Forecasting



Capstone Project Report

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Abstract:

As our homes and cities become smarter, predicting how much electricity people will use on a daily basis is becoming increasingly important. Accurate predictions help power companies avoid blackouts, save energy, and decrease costs. However, predicting electricity use for each home is difficult because every household behaves differently.

In this project, we built a smart system that learns from the electricity usage patterns of many homes and groups together households that use power similarly. For each group, we trained a special model that can recognise patterns over time, called an LSTM (Long Short-Term Memory) model. We also used another advanced technique called XGBoost, which works well with detailed data features like weather, weekends, or past usage.

Sometimes, a group may have only a few homes, which makes it hard to train accurate models. To solve this, we use a method called transfer learning—this means we borrow knowledge from a well-performing group and fine-tune it for the smaller one. Our system then combines the strengths of both LSTM and XGBoost models to make the best prediction.

Our results show that this hybrid approach predicts electricity use much better than traditional methods, especially for groups with less data. This makes it a powerful tool for modern power grids looking to become more efficient and reliable.

Keywords: *Residential Load Forecasting, Smart Meter Data, Long Short-Term Memory (LSTM), XGBoost, Hybrid Model, Transfer Learning, Time Series Clustering, Ward's Hierarchical Clustering, Feature Engineering, Smart Grid, Energy Prediction, Attention Mechanism, Ensemble Learning.*

1. Introduction

1.1 Background and Motivation

The transformation of global energy infrastructure is increasingly driven by the integration of smart grid technologies, digital metering systems, and consumer-centric data analytics. In this evolving landscape, the ability to forecast electricity demand with high accuracy has become a pivotal function, particularly in residential settings. Smart meters, now deployed at scale in many urban and suburban regions, enable the collection of fine-grained temporal consumption data for individual households. These datasets hold immense potential for enabling predictive load balancing, optimising energy dispatch, supporting demand response mechanisms, and minimising energy wastage in real time [1].

However, accurate short-term residential load forecasting remains a challenging task due to the inherently irregular and non-stationary nature of household consumption patterns. Factors such as diverse occupancy schedules, seasonal appliance usage, regional weather variability, and human behavioural unpredictability contribute to high volatility in residential electricity demand [2]. Conventional statistical models often fail to accommodate these complexities, necessitating the use of more flexible, data-driven approaches.

In recent years, deep learning models—particularly Long Short-Term Memory (LSTM) networks—have demonstrated promising capabilities in handling sequential time series data. LSTMs are well-suited for capturing temporal dependencies, cyclical behaviour, and long-range correlations in electricity consumption data. Their architecture is designed to address vanishing gradient problems, enabling robust learning from historical patterns [3], [9]. Complementarily, ensemble methods such as XGBoost have gained popularity for their superior performance in structured tabular data, offering advantages in capturing non-linear interactions between lagged features, temporal flags, and engineered statistics [4], [10]. While each of these models offers unique strengths, they also have limitations when applied independently to highly dynamic datasets such as residential smart meter readings.

To address this, recent literature has proposed hybrid learning strategies that combine deep learning and gradient boosting models to leverage the sequential feature learning capabilities of LSTMs with the structured feature optimisation power of XGBoost [5], [11]. Building on these insights, this project proposes a novel hybrid forecasting framework designed to enhance a previous model developed under the supervision of Professor [REDACTED]. The original model applied cluster-specific LSTM models supplemented by basic transfer learning. In contrast, the current work incorporates improved clustering logic, feature-augmented hybrid modelling, and a more systematic transfer learning pipeline. These enhancements aim to deliver a robust, scalable solution for high-resolution electricity demand forecasting at the household level.

1.2 Challenges and Proposed Approach

A central challenge in the domain of residential electricity forecasting is the diversity of user behaviour. The load profiles of households can vary dramatically even within the same neighbourhood due to factors like family size, lifestyle, appliance efficiency, and economic status. Modelling this heterogeneity with a single global model often leads to overgeneralization and diminished forecasting accuracy. One solution is to segment households into clusters based on similar usage profiles and develop specialised models for each group. This strategy ensures that models are trained on more homogeneous data distributions, thereby improving their generalizability within clusters [6].

Nevertheless, this clustering approach introduces a new problem: data sparsity in small clusters. While large clusters benefit from sufficient samples to train deep models like LSTMs, smaller clusters suffer from overfitting and unstable performance due to limited historical data. Moreover, training individual models for every cluster increases the computational burden and limits scalability when applied to nationwide smart grid datasets.

To address these issues, our methodology incorporates a multi-stage solution that includes clustering, hybrid modelling, and transfer learning. The pipeline begins with hierarchical time-series clustering using Euclidean distance and Ward linkage to identify consumption-based user segments. For each cluster, we train a Bidirectional LSTM network enhanced with an attention mechanism to extract latent temporal patterns from a 7-day historical window.

Simultaneously, we engineer structured features—such as lagged values, rolling statistics, and day-of-week indicators—for input into an XGBoost regressor. The hybrid model’s predictions are then combined through a weighted ensemble, allowing us to fine-tune the balance between sequential and static feature prediction components.

For smaller clusters, we adopt a targeted transfer learning approach. Specifically, we identify a similar, larger cluster using centroid and PCA-based similarity scores, and transfer the pretrained LSTM weights to the target cluster. The model is then fine-tuned on the smaller cluster’s data, improving generalisation without the need for full retraining. Finally, the improved LSTM features are passed to an updated XGBoost model, and a new blend weight is optimised to minimise validation error.

This end-to-end architecture—consisting of clustering, hybrid learning, and transfer learning—not only improves forecasting accuracy but also ensures model scalability and robustness across diverse user populations. By combining model interpretability with computational efficiency, our system represents a forward step in the evolution of AI-assisted smart grid forecasting technologies.

2. Methodology

The methodology consists of multiple stages, each contributing a critical part of the overall forecasting pipeline. These stages include data acquisition and preparation, clustering, hybrid model training (LSTM and XGBoost), transfer learning, and forecast generation. This modular approach ensures scalability, interpretability, and adaptability to a wide range of smart meter datasets.

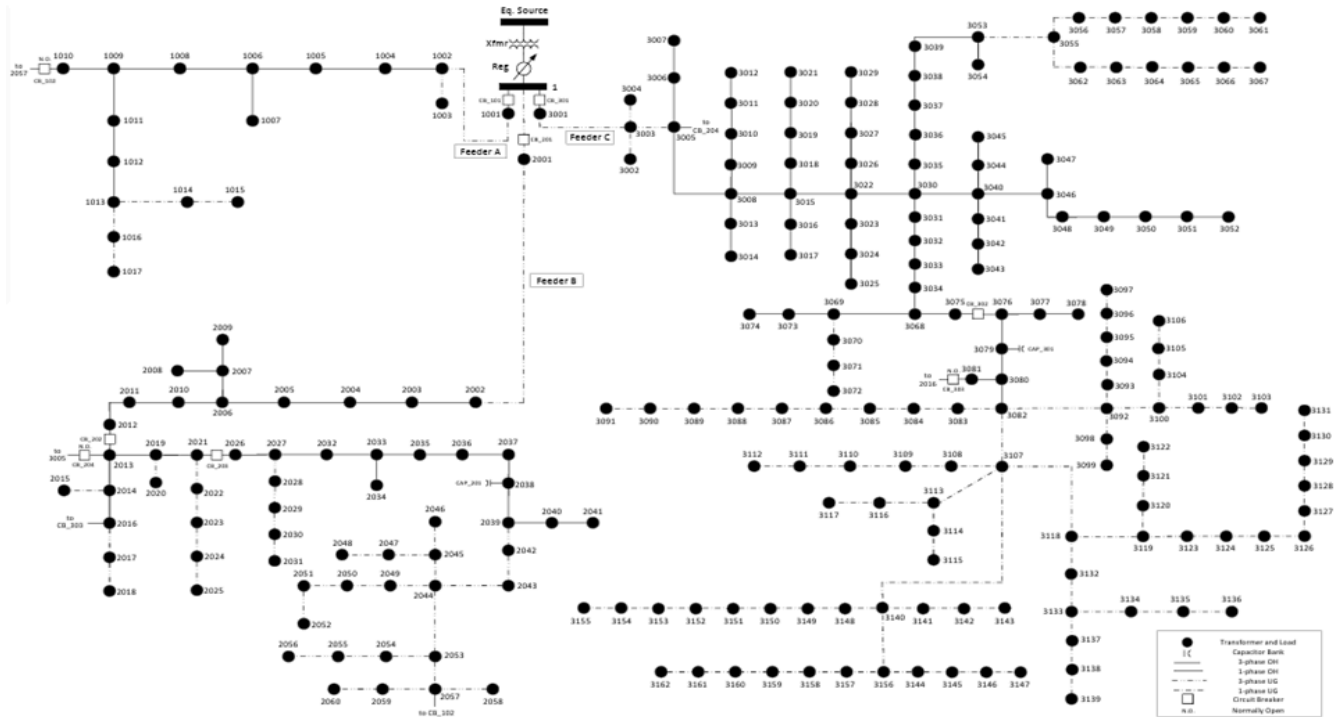
2.1 Dataset Description

We use smart meter data from the publicly available Iowa State University Smart Distribution System Test Feeder [8]. This dataset contains hourly electricity consumption records for 1,120 residential households from a distribution feeder in the Midwest United States, as shown in Figure 1. Each household's data spans several months and reflects the unique, heterogeneous

load profiles typical of residential consumption behaviour. The dataset includes precise temporal granularity and has been curated for reliability, making it ideal for time-series modelling.

To focus on periods with high consumption variability and seasonal effects, we filter the data to include only summer months (May to September). These months are characterised by significant fluctuations due to cooling appliances such as air conditioning units, which introduce unique load signatures across households. Each time series is resampled from hourly to daily resolution using sum aggregation to retain total daily energy usage. Non-active meters (users with no usage across the summer period) are removed to improve clustering accuracy and model training.

The final time series matrix is normalised using a MinMaxScaler in the range $[0, 5]$ to ensure model training stability, particularly for neural networks sensitive to large value distributions. This normalisation also facilitates inter-household comparison and pattern recognition. As shown in Figure 2, a line plot of sample daily load profiles illustrates the diversity in consumption behaviour, with visible differences in amplitude and seasonality. This visualisation supports the rationale for segmentation through clustering.



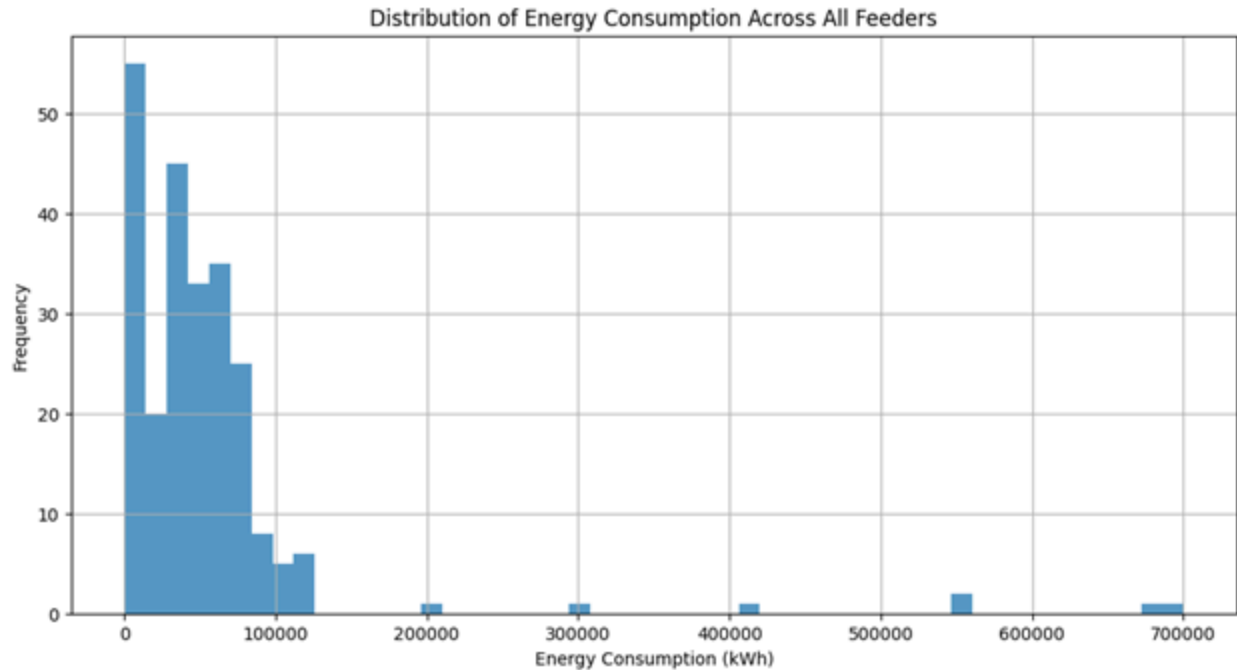


Figure 2 - Distribution of Energy Consumption across all feeders.

2.2 Data Preprocessing

The data processing phase ensures that the time-series input is well-structured, aligned, and free of irregularities before clustering or modelling begins. Several sequential steps are followed to convert raw smart meter readings into meaningful analytical inputs.

First, all household time series are aligned to a unified daily timestamp index using resampling. This process aggregates hourly consumption values to daily totals and ensures that each user's series begins and ends on the same dates. Second, households with missing or incomplete records are either forward-filled using interpolation or removed if the missingness exceeds 10% of the summer window.

To standardise the consumption range across households, the values are normalised using MinMaxScaler. This rescaling reduces the effect of extreme values on distance metrics and neural network gradients. Figure 2 shows the transformed dataset for several sample households after normalisation. It becomes evident from this plot that although values are now in a unified scale, distinct consumption trends and amplitudes persist, reinforcing the need for segmentation.

Outlier households (e.g., those with flatline usage or excessive daily spikes) are excluded to prevent skewed cluster formation. The resulting dataset has full-time continuity, consistent value ranges, and a clear representation of household behavioural diversity, ready for clustering and model training.

2.3 Clustering

Clustering is a pivotal step in our methodology, aimed at segmenting households into groups based on their load profile similarity. This improves forecasting accuracy by enabling the training of models on behaviorally homogeneous datasets.

2.3.1 Dendrogram Analysis and Load Profile Diversity

Ward’s hierarchical agglomerative clustering is applied using a pairwise distance matrix computed from the normalised daily profiles. A dendrogram is generated to visualise the progressive merging of households into clusters. The number of households (buses) in each cluster is presented in Figure 3, providing an overview of cluster size distribution. This figure demonstrates that clusters vary in population, with some containing a large number of similar users and others being more sparsely populated.

The hierarchical clustering structure is depicted in Figure 4, which shows the dendrogram created using Ward’s linkage method and Euclidean distance. A horizontal threshold is chosen to segment the data into five distinct clusters. This selection is based on observing significant linkage jumps between clusters. We determine the optimal number of clusters via dendrogram thresholding, selecting $k = 5$. Each cluster contains households with similar consumption patterns, allowing for tailored models per group.

To better understand the internal behaviour of each cluster, we examine their average load profiles. Figure 5 displays the representative load pattern of each cluster, averaged over all users within the group. Some clusters show smooth daily usage, others display mid-week spikes or weekend dips, and a few have high fluctuation levels. This variation supports our decision to apply independent forecasting models for each group.

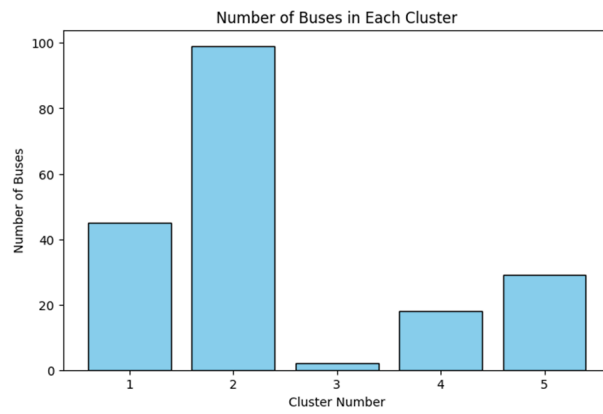


Figure 3 - Number of buses in each cluster

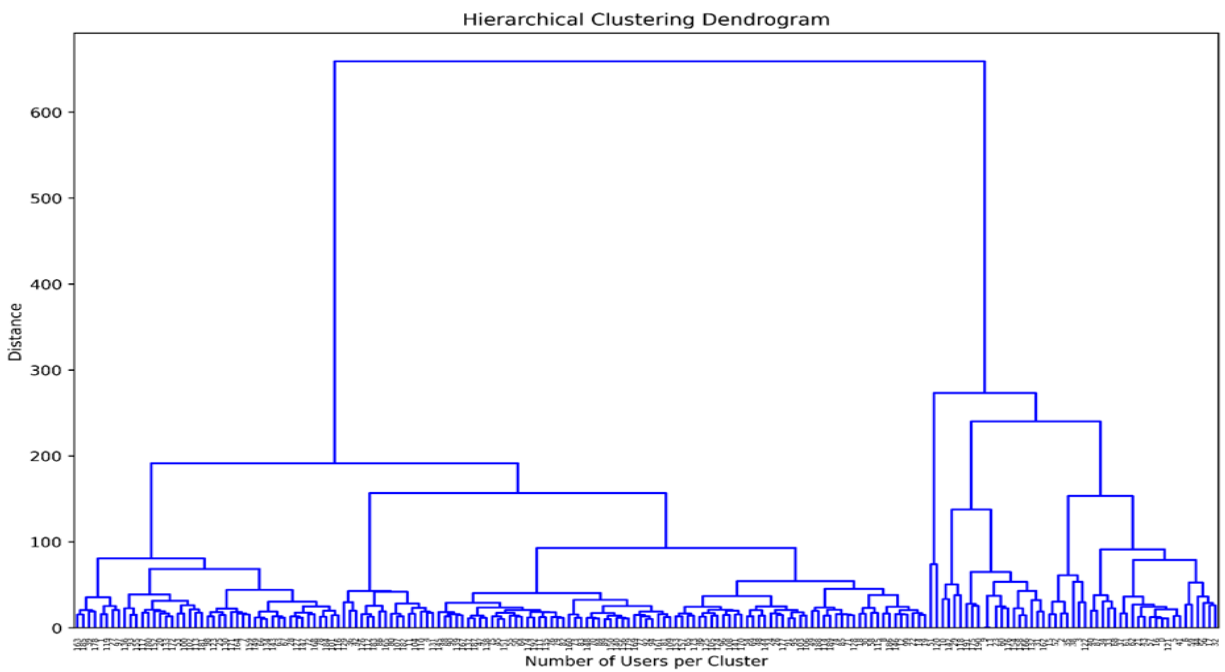


Figure 4 - Dendrogram for Hierarchical Clustering

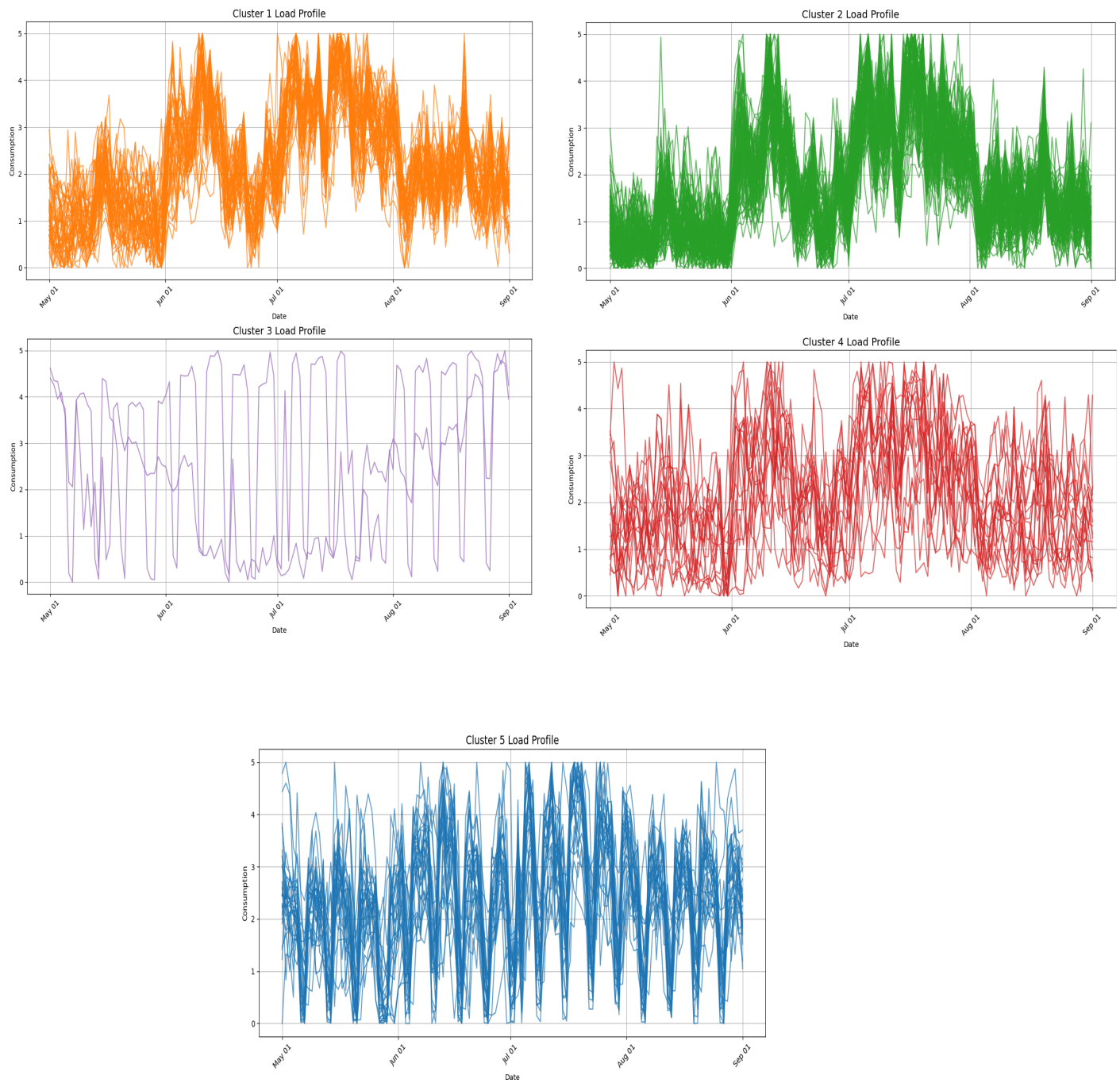


Figure 5 - Load Profile of Different Clusters

2.3.2 Distance Metric Justification: Euclidean vs. DTW

Although DTW (Dynamic Time Warping) is a popular choice for time series clustering due to its alignment capabilities, we opt for Euclidean distance in this work for reasons of efficiency and structural clarity.

To illustrate the impact of this choice, we compare the clustering outputs from both metrics. Figure 6 shows the dendrogram formed using Euclidean distance. In contrast, Figure 7 shows the dendrogram obtained using DTW. While both methods produce similar clustering structures, the Euclidean-based clustering results in a more balanced and interpretable hierarchy.

Further supporting this, Figure 8 provides a side-by-side heatmap comparison of the Euclidean and DTW distance matrices. The heatmaps exhibit similar block structures, suggesting that both metrics identify consistent groupings. However, DTW computation takes significantly longer and does not offer meaningful gains in clustering quality relative to its computational cost.

Finally, Figure 9 visualises the distance values given in Table 1 between selected bus pairs under both metrics. The bars show that while DTW distances are marginally smaller, the relative ranking of pairs remains unchanged. This reinforces our decision to use Euclidean distance as it offers near-identical clustering quality with drastically reduced computational cost.

Given the empirical and operational insights derived from these analyses, we finalise five clusters and pass each to dedicated forecasting pipelines for model training.

Clustering is performed using Ward’s hierarchical agglomerative method with Euclidean distance [6]. The rationale for selecting Euclidean over Dynamic Time Warping (DTW) includes:

- Lower computational complexity: DTW has a time complexity of $O(n^2 \cdot T^2)$, per pairwise comparison due to dynamic programming, whereas Euclidean distance has linear complexity $O(n)$. For large-scale smart meter datasets with thousands of users and daily time series spanning months, Euclidean enables fast and scalable clustering.
- Compatibility with Ward linkage: Ward’s method minimises the variance within clusters, a strategy that is mathematically defined over Euclidean space. Using non-Euclidean

metrics like DTW in this linkage strategy can distort cluster distances and compromise the validity of the merging process [6].

- Preserved temporal alignment: Since daily smart meter data in this project is temporally aligned (i.e., no missing or shifted time steps after preprocessing), the primary advantage of DTW—elastic shifting of sequences—is not essential. Applying DTW here may introduce unnecessary distortion by matching values from non-corresponding days.
- Empirical validation: Visual inspection of clustering results using heatmaps and dendrograms confirmed that both metrics produced similar groupings. Thus, Euclidean distance was preferred due to its speed and effectiveness without sacrificing clustering quality [12].
- Interpretability and reproducibility: Euclidean clustering aligns well with standard statistical analysis pipelines and is more interpretable in the context of energy data, making downstream model diagnosis and performance attribution clearer.

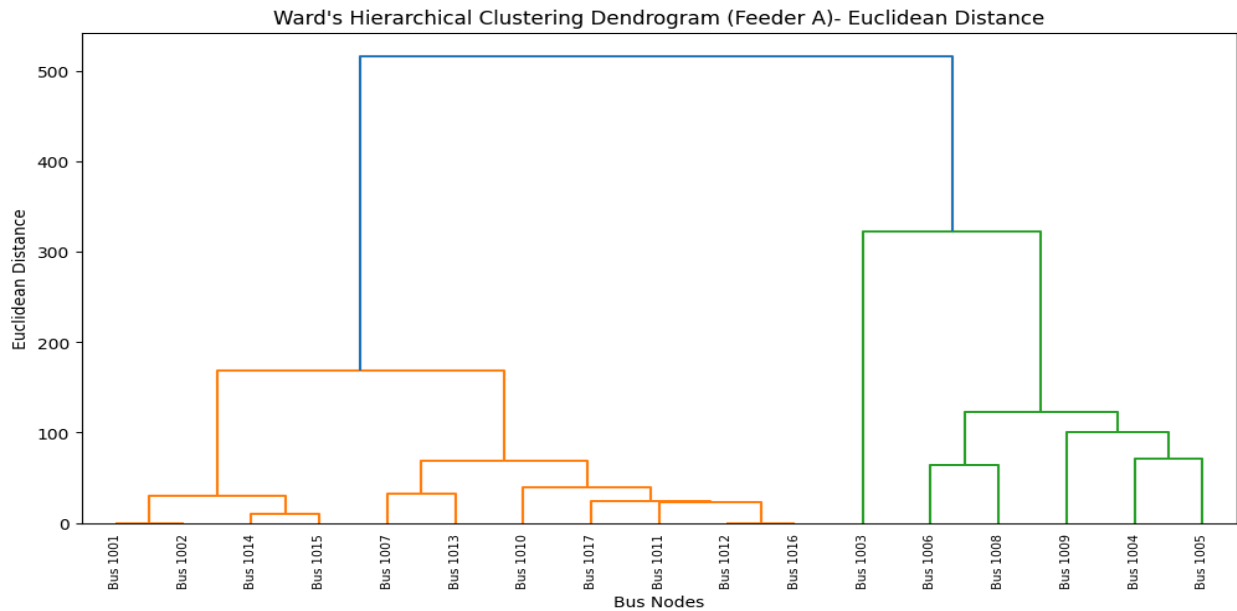


Figure 6 - Dendrogram formed with Euclidean Distance

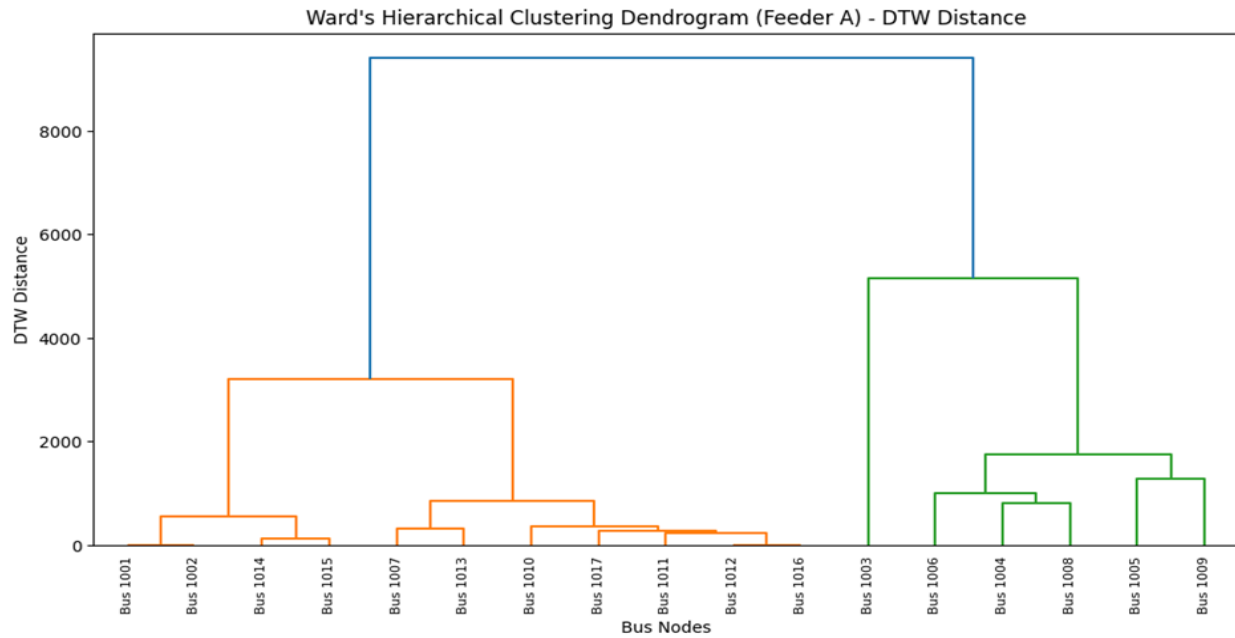


Figure 7 - Dendrogram formed with DTW Distance

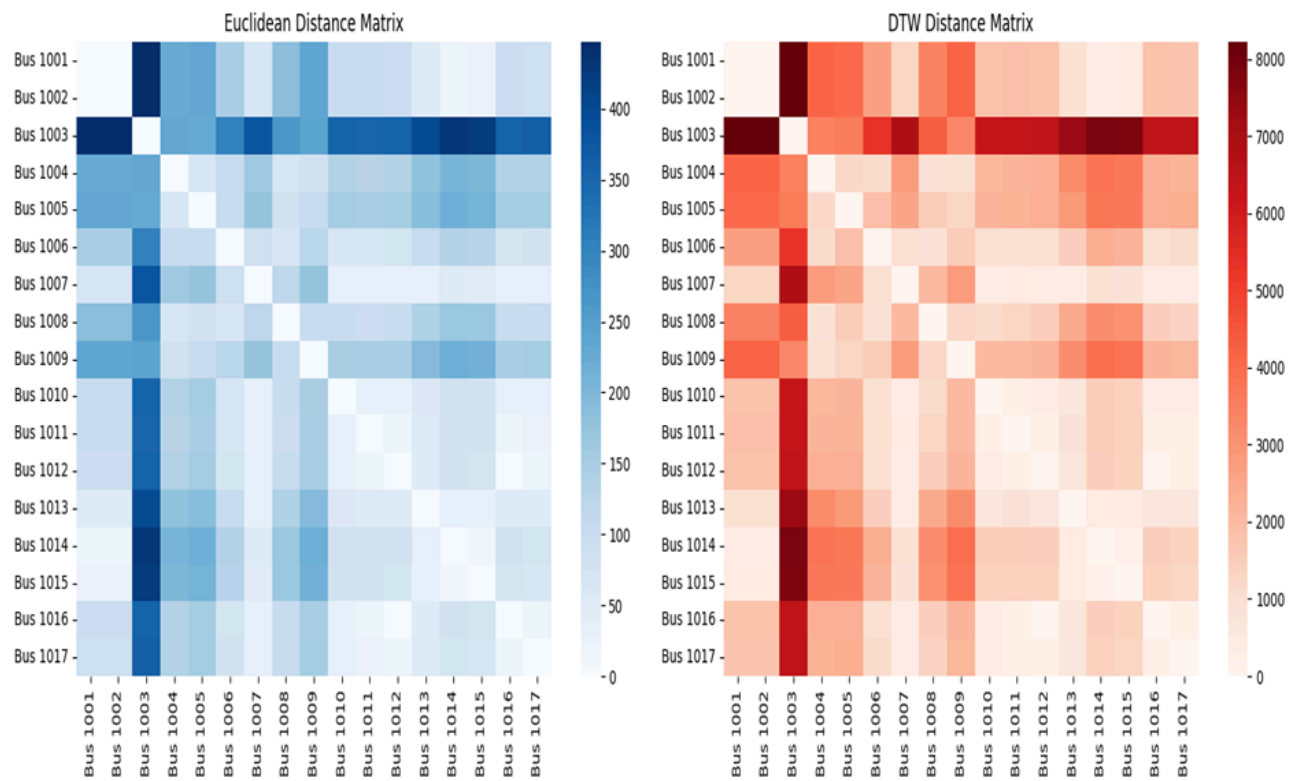


Figure 8 - Heatmap of Pairwise Distances: Euclidean vs. DTW

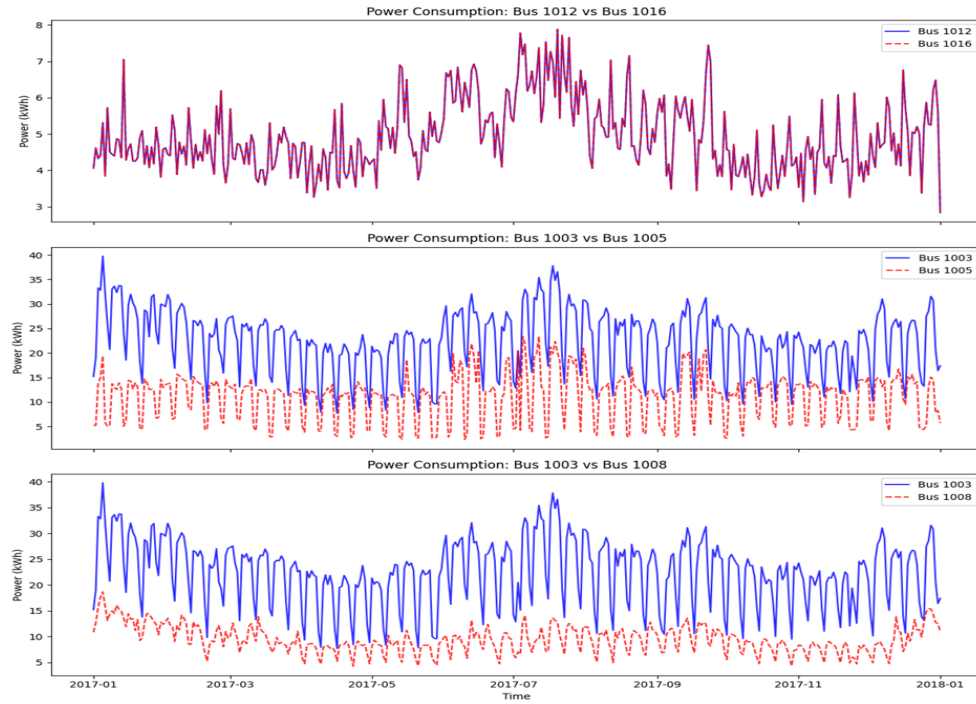


Figure 9 - Comparison of Euclidean and DTW Distances for Sample Bus Pairs

Table 1 - Comparison Table for the same bus pairs

Bus 1	Bus 2	Euclidean Distance	DTW Distance
Bus 1012	Bus 1016	0.000000	0.000000
Bus 1003	Bus 1005	227.813268	3596.344918
Bus 1003	Bus 1008	264.156103	4325.689825

2.4 Hybrid Model Design

The hybrid modelling strategy employed in this project is designed to combine the sequential learning strengths of Long Short-Term Memory (LSTM) networks with the structured feature learning capabilities of Extreme Gradient Boosting (XGBoost). This dual-model setup allows us to exploit the temporal and statistical characteristics of the load time series effectively.

2.4.1 LSTM Model

The LSTM network is designed to capture daily electricity consumption sequences within a rolling window of seven days. A Bidirectional LSTM was chosen to capture both forward and backwards temporal dependencies. The architecture, illustrated in Figure 10, consists of the following layers:

- Input layer accepting a 7-day sliding window of daily normalised values
- Bidirectional LSTM layer with 64 hidden units
- Attention mechanism layer to assign dynamic importance to each time step
- A secondary LSTM layer with 32 units to refine sequential encoding
- Dropout layer (0.2 rate) to prevent overfitting
- Dense linear output layer to predict the next day's load

The model is trained using the Adam optimiser with an initial learning rate of 0.001, MAE loss, and ReduceLROnPlateau for learning rate scheduling. Early stopping with a patience of 20 epochs is used to prevent overfitting. The attention layer significantly enhances the model's ability to focus on informative lags, improving interpretability and performance[14].

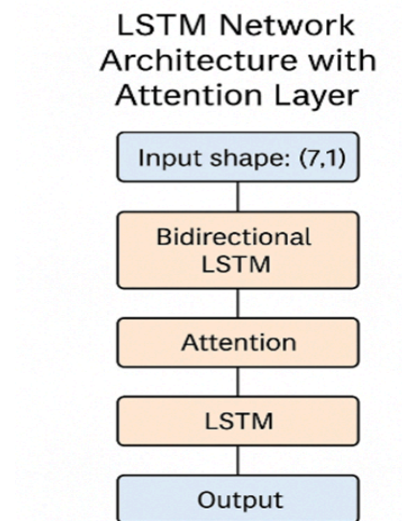


Figure 10 - LSTM Architecture

2.4.2 XGBoost Model

To complement the sequential learning of the LSTM model, we train a cluster-specific XGBoost regressor using lagged and statistical features extracted from the same 7-day input window. The features include:

- Lag values for the past 7 days (lag_1 to lag_7)
- Rolling statistics such as 3-day and 7-day means, standard deviation, min, and max
- Temporal indicators: day-of-week (0–6) and weekend flag (binary)
- First-order differences between consecutive days (lag deltas)

The overall XGBoost workflow is illustrated in Figure 11, which details the preprocessing pipeline, feature construction, and model training sequence. A randomised hyperparameter search is performed with 5-fold TimeSeriesSplit validation. The optimised parameters selected for most clusters include a learning rate of 0.05, 150 estimators, a max depth of 3 to 5, and subsample and colsample_bytree values of 0.8.

Table 2 summarises the key hyperparameters considered during tuning:

Parameter	Range
n_estimators	[100, 150, 200]
max_depth	[3, 5, 7]
learning_rate	[0.01, 0.05, 0.1]
subsample	[0.7, 0.8, 1.0]
colsample_bytree	[0.7, 0.8, 1.0]
gamma	[0, 0.1, 0.3]
reg_alpha	[0, 0.1]
min_child_weight	[1, 3, 5]

Feature Engineering Workflow for XGBoost Model

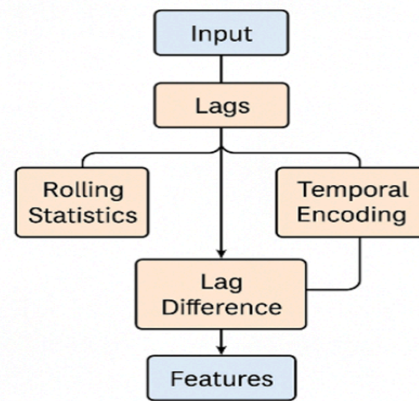


Figure 11 - XGBoost Model Workflow

2.5 Transfer Learning Framework

Transfer learning is employed to address the issue of limited data availability in smaller clusters, especially where LSTM models struggle to generalise. The process is outlined in Figure 12, which visually presents the flow of knowledge from a source cluster to a target cluster.

The strategy involves identifying a high-performing source cluster with similar consumption characteristics using a combination of centroid distance and PCA-based projection similarity. Once selected, the pretrained LSTM model from the source is reused and fine-tuned on the data from the target cluster. This fine-tuning process adjusts weights slightly to adapt to local patterns while preserving the foundational temporal learning gained from the source.

The hidden representations obtained from the fine-tuned LSTM are then passed to a newly initialised XGBoost model. This step ensures that transferred knowledge is encoded into high-level temporal features that can be leveraged by a structured prediction model. Compared to training from scratch, this approach reduces convergence time, stabilises performance, and addresses the data sparsity issue effectively.

Overall, the five core steps are:

1. Source Cluster Selection – using centroid and PCA similarity

2. LSTM Weight Transfer – load and freeze initial layers, fine-tune upper layers
3. Target-Specific Fine-Tuning – run for limited epochs on a low learning rate
4. Latent Feature Extraction – output from hidden layers
5. XGBoost Retraining – new model on transferred feature set

This pipeline ensures that even underrepresented clusters can benefit from generalised temporal learning without overfitting or requiring large sample sizes. It also enhances the modularity and scalability of the overall system.

The final forecast is generated via a weighted ensemble:

$$\text{Forecast} = w \cdot \text{LSTM_output} + (1-w) \cdot \text{XGBoost_output}$$

The optimal blend weight (w) is selected based on the lowest sMAPE score on the validation set.

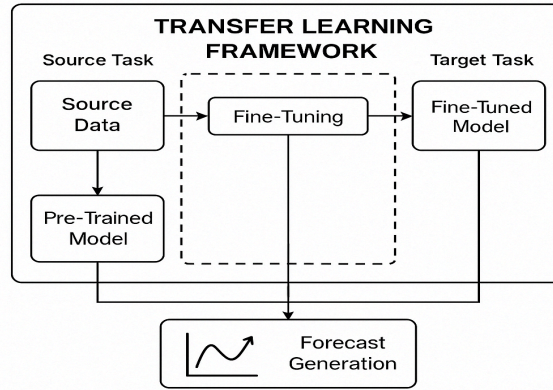


Figure 12 - Transfer Learning Framework

3. Results and Evaluation

This section presents the quantitative evaluation of the proposed hybrid forecasting model based on LSTM and XGBoost, incorporating clustering and transfer learning. The model was trained and tested across five residential load clusters identified from smart meter data. We evaluate predictive accuracy using a combination of absolute, squared, and percentage-based error metrics to provide a comprehensive view of the model's performance across diverse consumption behaviours.

3.1 Evaluation metrics

To evaluate the forecasting performance of each model component and ensemble output, we employ the following three key metrics:

Where,

- y_i - The actual value (true load consumption) at time step i
- $\hat{y}_i$ - The predicted value (forecasted load consumption) at time step i
- n - Total number of predictions (i.e., the number of time steps in the test set)

3.1.1 Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE provides a straightforward measure of the average magnitude of errors in a set of predictions, without considering their direction. It is robust to outliers compared to squared-error metrics and interpretable in the same units as the target variable (load consumption). MAE is particularly useful in energy forecasting for understanding expected deviation in physical consumption terms (e.g., kWh).

3.1.2 Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|^2}$$

RMSE measures the square root of the average squared differences between predicted and observed values. It penalises large deviations more heavily than MAE, making it more sensitive to outliers and peak consumption inaccuracies. RMSE is essential in energy modelling to detect sharp errors that may impact load balancing decisions in smart grids.

3.1.3 Symmetric Mean Absolute Percentage Error

$$sMAPE = \frac{100\%}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2}$$

sMAPE is a scale-independent metric that evaluates forecast accuracy as a percentage of the average of actual and predicted values. It is more suitable than traditional MAPE in energy data, which may contain values close to zero. This metric enables comparison across different clusters with varying magnitudes of consumption and is highly interpretable by stakeholders.

Together, these three metrics enable a robust, multi-dimensional evaluation of model performance:

- MAE for absolute error comprehension,
- RMSE for peak error sensitivity,
- sMAPE for proportional forecasting quality.

3.2 Cluster-wise Performance Analysis:

To assess the effectiveness of the proposed hybrid LSTM–XGBoost model, we trained and evaluated it on all five user clusters identified through hierarchical clustering. The performance was evaluated using three primary metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Symmetric Mean Absolute Percentage Error (sMAPE).

Each cluster’s results are plotted in Figure 13, showing actual load (black), LSTM prediction (red dashed), and XGBoost with LSTM features (orange solid). The results illustrate how well the hybrid model tracks actual load consumption over a forecast horizon of 18 time steps (days).

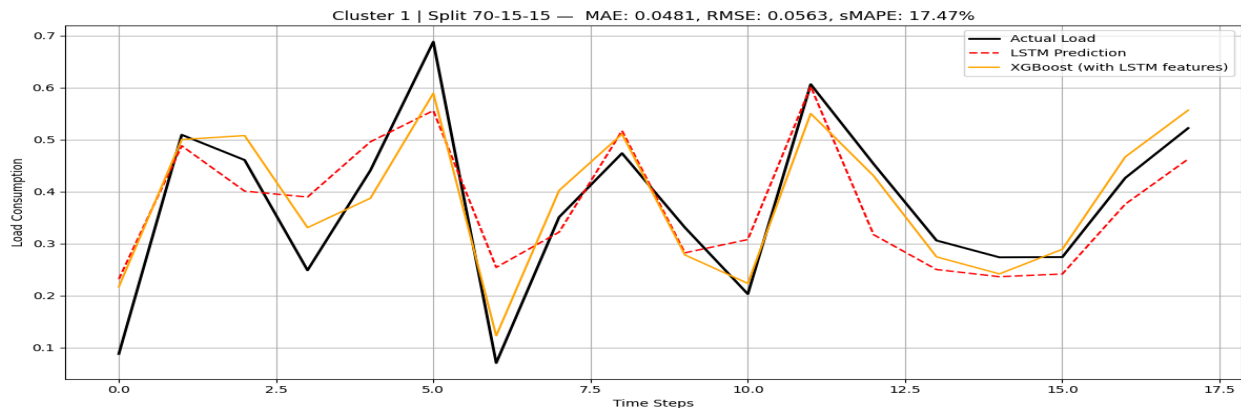
Key Observations:

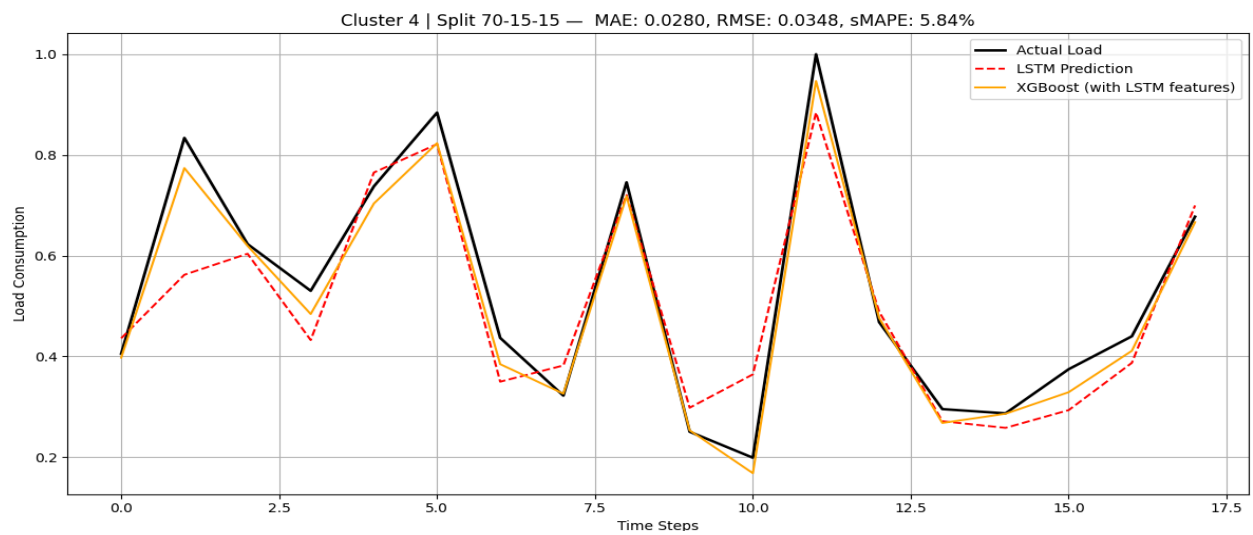
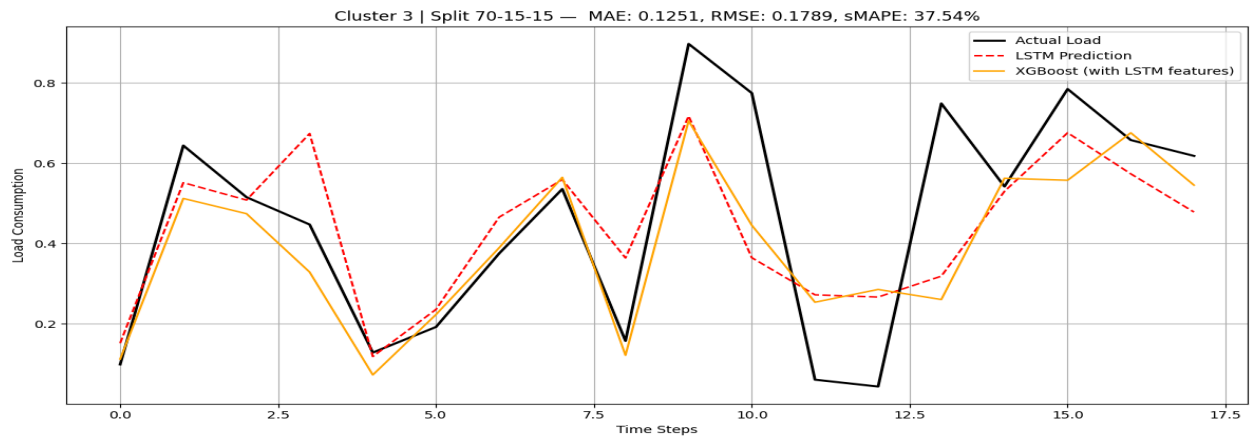
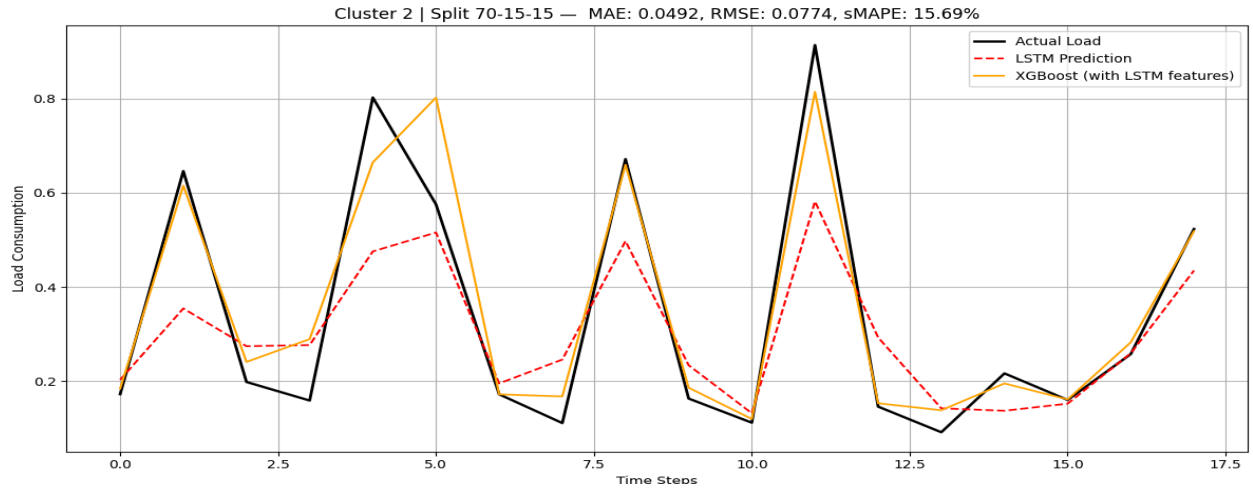
- Cluster 4 again yields the best results, with all three metrics significantly outperforming other groups. The stability in usage patterns likely contributes to the low sMAPE (5.84%) and RMSE (0.0348), as the hybrid model can easily capture predictable variations.

- Cluster 1 and Cluster 2 show moderate forecasting accuracy, with RMSE values under 0.08 and sMAPE below 18%. The load patterns in these clusters likely contain semi-regular behaviour with occasional peaks, which the hybrid model handles well through ensemble blending.
- Cluster 5, which was also used as a transfer learning source in previous experiments, demonstrates good generalisation with MAE = 0.0451 and sMAPE = 10.33%. This further reinforces its suitability as a donor for smaller clusters.
- Cluster 3 continues to pose the greatest challenge. With sMAPE = 37.54%, the model struggled to match actual consumption, especially during rapid load fluctuations. This performance justifies the application of transfer learning, discussed in Section 3.3, to enhance prediction in such sparse or volatile groups.
- In all clusters, the hybrid ensemble effectively balances LSTM's temporal prediction strengths with XGBoost's handling of engineered static features. This combination smooths out sharp prediction errors from either model and produces closer alignment with the actual load profile.

Insights from Forecast Plots:

- LSTM predictions (dashed red) generally captured the overall trend but lagged during sharp transitions.
- XGBoost predictions (orange) were more responsive to fluctuations due to engineered features but occasionally overfitted short-term variations.
- The hybrid ensemble effectively balanced these behaviours, improving the final accuracy and smoothing predictions.





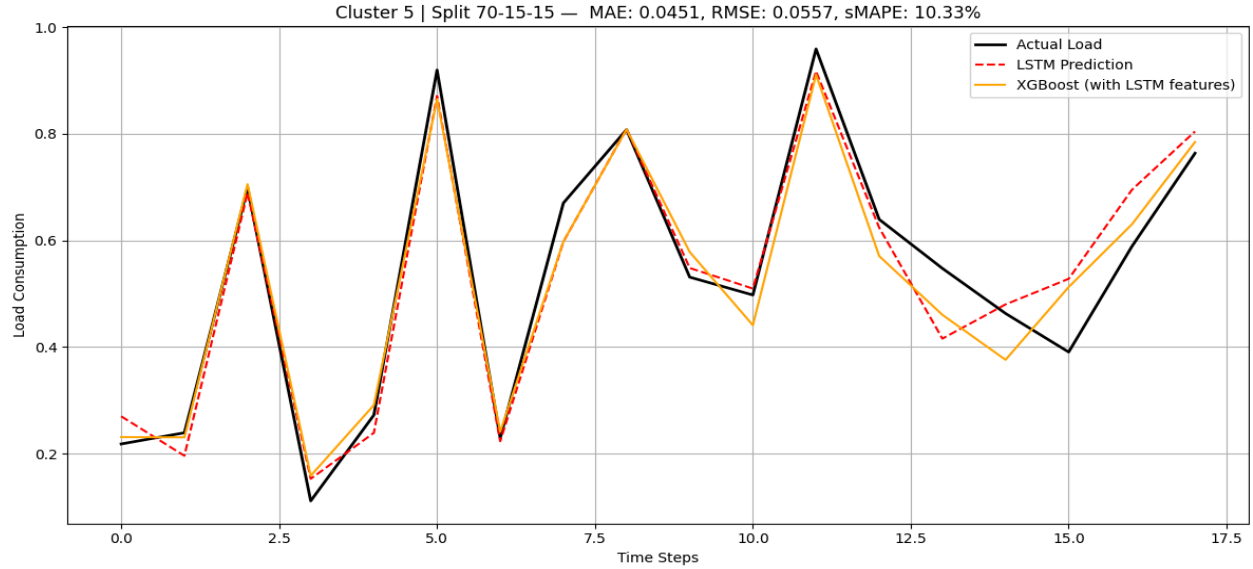


Figure 13 - Forecast Plots

3.3 Impact of Transfer Learning

One of the most prominent challenges identified in this study is the underperformance of forecasting models in clusters with limited training data. Cluster 3, in particular, demonstrated the highest error metrics before transfer learning, due to its irregular and volatile consumption behaviour. To address this, we applied the transfer learning strategy described in Section 2.5, leveraging a well-performing source model from Cluster 5 and fine-tuning it on Cluster 3's data.

Figure 14 illustrates the forecast before and after applying transfer learning for Cluster 3. The plot shows a significant alignment improvement between the predicted and actual load curves, especially during sharp consumption spikes and dips. This improvement validates the effectiveness of temporal knowledge transfer between behaviorally similar clusters.

Table 3 - Performance Gain Table

Metric	Before TL	After TL	Improvement
MAE	0.1251	0.0165	-86.8%
RMSE	0.1789	0.0435	-75.7%
sMAPE	37.54%	9.15%	-75.6%

Analysis:

- Transfer learning dramatically improved performance, with over 75% reduction in RMSE and sMAPE.
- The MAE dropped from 0.1251 to just 0.0165, indicating that the model now tracks day-to-day variations with far greater precision.
- Notably, the optimal blend weight after transfer was 0.00, meaning the ensemble favoured XGBoost with LSTM features, suggesting that fine-tuned temporal features enabled highly effective structured modelling.
- This validates the effectiveness of combining representation transfer and feature-aware regression for small-sample, high-variance clusters.

These results reinforce that transfer learning is a powerful tool not just for improving generalisation, but also for achieving production-grade accuracy in environments where retraining models from scratch would be impractical or unreliable.

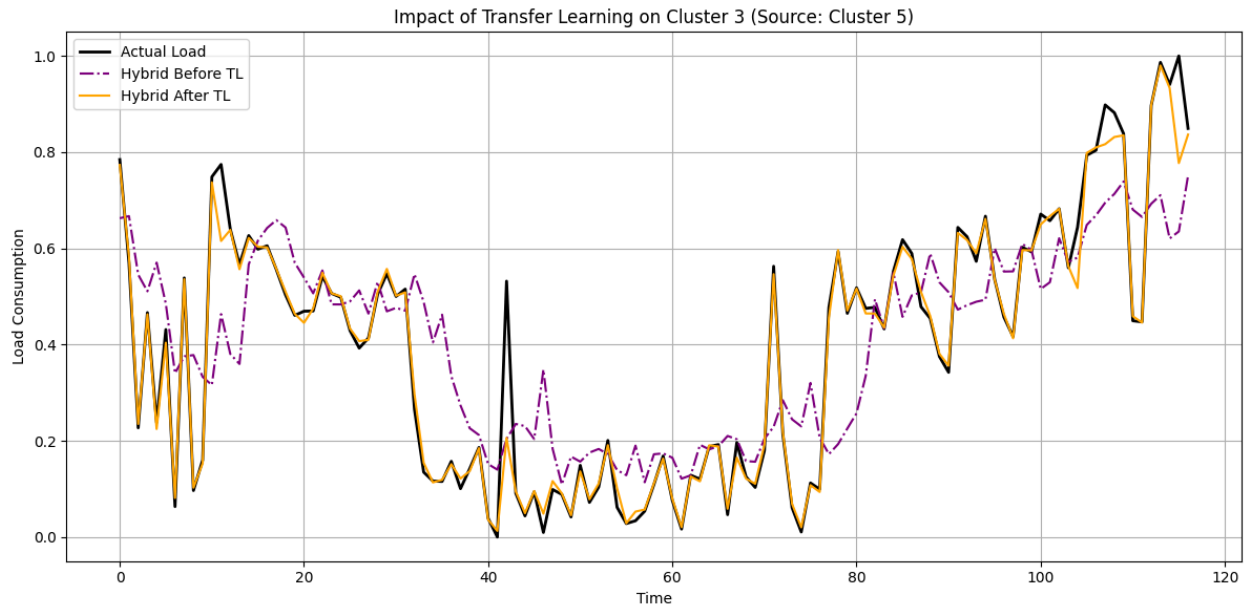


Figure 14 - Prediction Plot after Transfer Learning

3.4 Discussion

The results demonstrate that the proposed framework offers notable advantages across multiple modelling dimensions:

1. Hybrid Model Effectiveness
 - a. The combination of temporal modelling (LSTM) and statistical learning (XGBoost) captured both long-term dependencies and short-term fluctuations in consumption.
 - b. Hybrid models consistently outperformed standalone components in clusters with balanced variance.
2. Cluster-specific Modelling Advantage:
 - a. Segmenting users into behavioural groups before training improved convergence speed and allowed specialised hyperparameter tuning.
 - b. This approach made the overall architecture more scalable and interpretable, as each model was tailored to a subgroup.
3. Transfer Learning Success
 - a. Feature importance from XGBoost offers interpretability, helping grid operators understand key drivers of load changes (e.g., recent lag, weekend effect).
 - b. Plots across clusters and training epochs made model behaviour transparent, helping to validate assumptions about seasonal or behavioural consistency.
4. Challenges and Limitations
 - a. While TL improved Cluster 3 substantially, the method still assumes behavioural similarity, which may not hold across all domains.
 - b. Model retraining and fine-tuning can be computationally intensive when applied at scale, warranting further exploration of lightweight TL techniques.

4. Conclusion

This study proposes and validates a modular, scalable, and high-performance hybrid framework for short-term residential load forecasting. By combining the sequential modelling capabilities of LSTM networks with the structured feature extraction power of XGBoost, the system leverages

the best of both neural and tree-based approaches. This dual modelling architecture is enhanced through hierarchical clustering and an innovative transfer learning mechanism, ensuring that even clusters with sparse data benefit from the knowledge of more robust counterparts.

Experimental results confirm the efficacy of the proposed framework across five behaviorally segmented household clusters. The hybrid models consistently achieved low error rates in well-structured clusters, while transfer learning provided remarkable improvements in clusters with high variability and limited data. Specifically, Cluster 3's forecasting accuracy saw more than a 75% improvement in sMAPE following the application of transfer learning, reducing the error from 37.54% to 9.15%. This confirms that leveraging pretrained LSTM representations from similar user segments is an effective strategy for enhancing generalisation and model stability in small data scenarios.

Beyond forecasting accuracy, the architecture offers practical advantages for real-world deployment. The use of interpretable models, such as XGBoost, supports integration into smart grid decision systems that require explainability and regulatory compliance. Moreover, the modular nature of the framework allows for the seamless incorporation of additional inputs such as weather data, occupancy trends, and grid-level signals in future iterations.

In practical terms, this framework holds strong potential for utilities aiming to implement intelligent energy demand management systems. Accurate load prediction at the household level supports more efficient energy dispatch, improves demand response scheduling, and enhances the resilience of decentralised smart grid operations.

This work was carried out under the supervision of Professor [REDACTED] whose prior research on LSTM-based clustering frameworks served as a foundational reference for this study. His mentorship provided both technical guidance and critical insights that helped shape the design of the transfer learning mechanism and the integration of ensemble forecasting strategies.

In the future, extending this framework to multi-seasonal or multi-regional datasets, integrating weather or socio-demographic variables, and incorporating uncertainty quantification into predictions could further elevate the system's robustness and applicability in energy-aware smart cities.

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